Introduction: Basics

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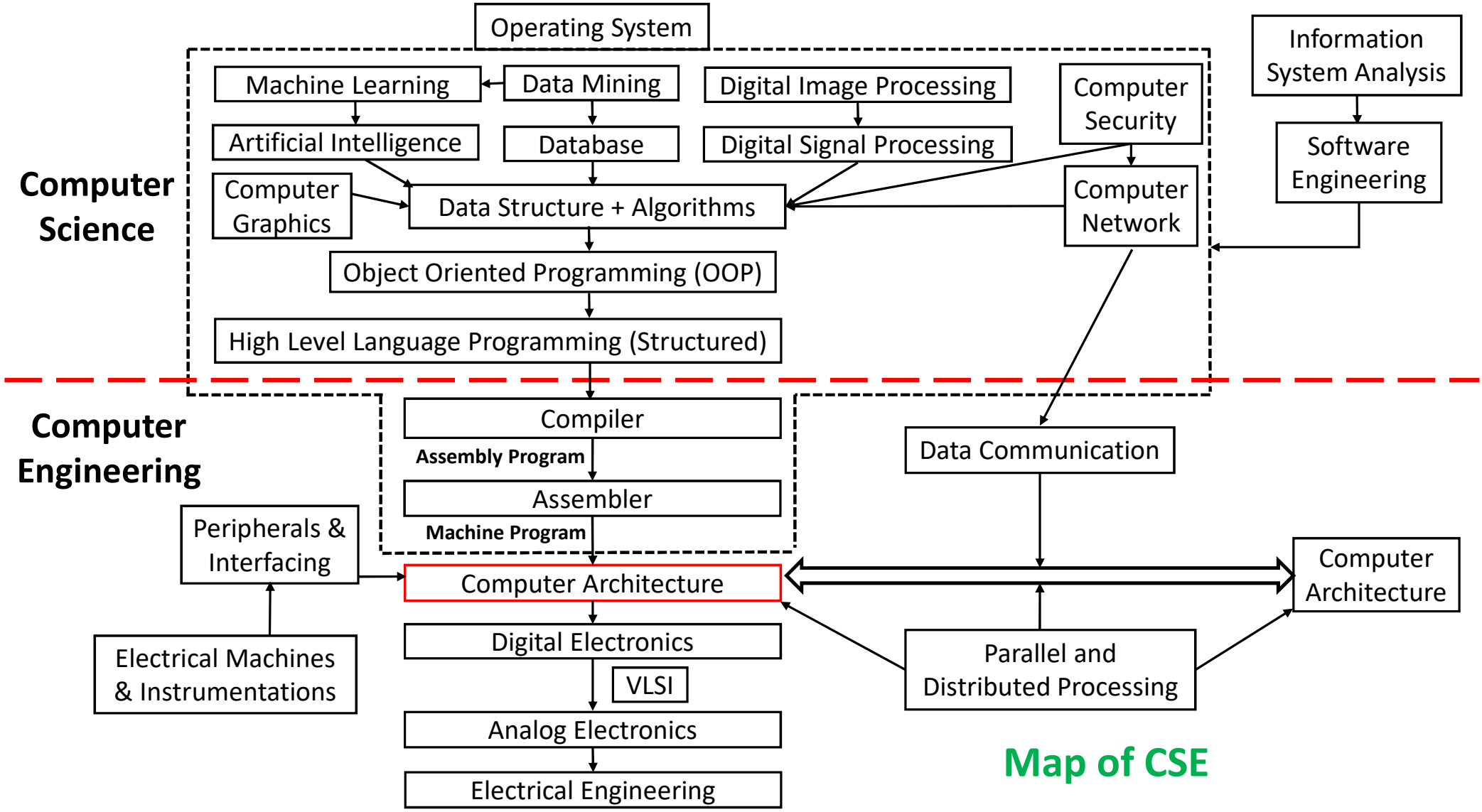
Importance of Computer Architecture

Computer Architecture:
In computer engineering, computer architecture is a set of rules and methods that describe the functionality, organization, and implementation of computer systems.

Computer architecture is the combination of microarchitecture and instruction set architecture.

Computer Organization (Microarchitecture):
In computer engineering, microarchitecture, also called computer organization and sometimes abbreviated as µarch or uarch, is the way a given instruction set architecture (ISA) is implemented in a particular processor.

A given ISA may be implemented with different microarchitectures. Implementations may vary due to different goals of a given design or due to shifts in technology.


Computer Science
Operating System
Machine Learning
Data Mining
Digital Image Processing
Computer Security
Information System Analysis
Artificial Intelligence
Database
Digital Signal Processing
Software Engineering
Computer Graphics
Data Structure + Algorithms
Computer Network
Object Oriented Programming (OOP)
High Level Language Programming (Structured)
Computer Engineering
Compiler
Assembly Program
Assembler
Machine Program
Data Communication
Peripherals & Interfacing
Computer Architecture
Computer Architecture
Electrical Machines & Instrumentations
Digital Electronics
VLSI
Parallel and Distributed Processing
Analog Electronics
Electrical Engineering

Map of CSE

Career Opportunities in Hardware (Semiconductor) Industry

intel

NVIDIA

arm

Qualcomm

AMD

MICROCHIP

SAMSUNG

Computer Engineering Jobs

Hardware Engineer
Arm
Budapest, HU

Hardware Engineer - College Graduate
Qualcomm
San Diego, CA, US

System Design Engineer - Hardware
NVIDIA
Santa Clara, CA, US

Hardware Design Engineer
Intel Corporation
Santa Clara, CA, US

ASIC Digital Design Engineer (Video)
Qualcomm
San Diego, CA, US

ASIC IP Verification Engineer (75900)
AMD
Orlando, FL, US

Relationship between Hardware and Software

Question: Can high level language program (Software) be run on computer processor (Hardware) directly?

Ans: Computer understands only program written in 0/1. It is called Machine Code/Program.

So, High level language program must be converted to Machine Language Program.

Conversion from High Level Program to Machine Code

```
int main()
{
    int a=10;
    a=a+5;
    return 0;
}
```

High Level Language

Compiler

```
call    ___main
mov DWORD PTR [esp+12], 10
add DWORD PTR [esp+12], 5
mov eax, 0
```

Assembly Language

Assembler

```
c7 44 24 0c 0a 00 00
83 44 24 0c 05
b8 00 00 00 00
```

Machine Language

Question: What kind of machine code does computer processor understand?

Ans: Every computer processor has something called **Instruction Set Architecture (ISA)** which defines machine code for specific instruction.
Compiler/Assembler must generate machine code based on ISA. ISA is the one that connects Hardware and Software together.

Question: How does computer processor itself run machine code?

Ans: Every modern processor is based on Von Neumann Model.

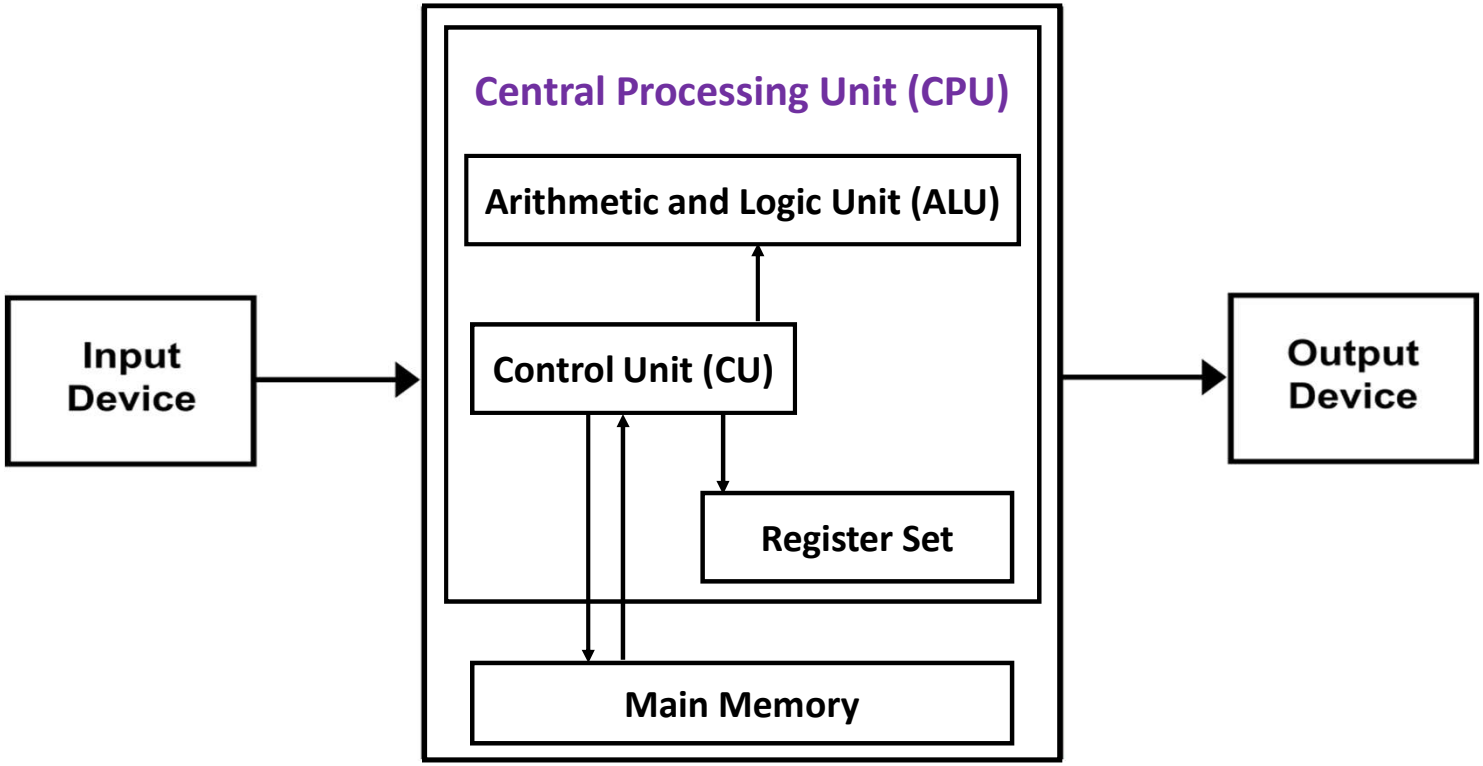


Figure: Von Neumann Model

1. Computer will fetch instruction from Main Memory (RAM).
2. Instruction will be decoded by control unit and will select registers and/or immediate values.
3. Data within registers and/or immediate values will be sent to Arithmetic and Logic Unit (ALU) to perform operations.
4. ALU will perform operation and result will be sent to the register to be written.
5. Control unit can send data from registers to Main memory.

Question: How are computer building blocks like ALU, CU, Register Set, Main Memory etc. made of?

Ans: They are made of digital electronics building block like AND/OR etc.

Question: How are digital electronics building block like AND/OR etc. made of?

Ans: They are made of analog electronics building block called transistor.

Question: How does a high level language program run on transistor inside of computer processor?

Answer: Connection between High Level Language Program and Transistor

Software

High Level Language Program

Compiler

Assembly Language Program

Assembler

Machine Language Program

```
int main(){int a=10; a=a+5; return 0;}
```

```
mov DWORD PTR [rbp-4], 10
add DWORD PTR [rbp-4], 5
```

```
c7 45 fc 0a 00 00 00
83 45 fc 05
```

`c7` is opcode for mov
`83` is opcode for add

From Intel ISA

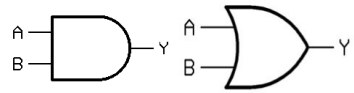
Instruction Set Architecture

Hardware

Computer Architecture

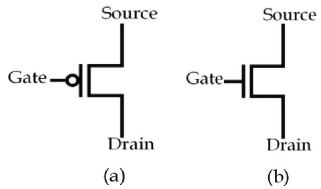
ALU | CU | Register Set | Main Memory

Digital Electronics



Adder | Multiplier | Flip-flop

Analog Electronics



Transistor
(Acts as a switch)

Physics (Law of Nature)

Relationship between Theory of Computation and Computer Architecture

Turing Machine

Turing Machine is the mathematical model of real-life computers.

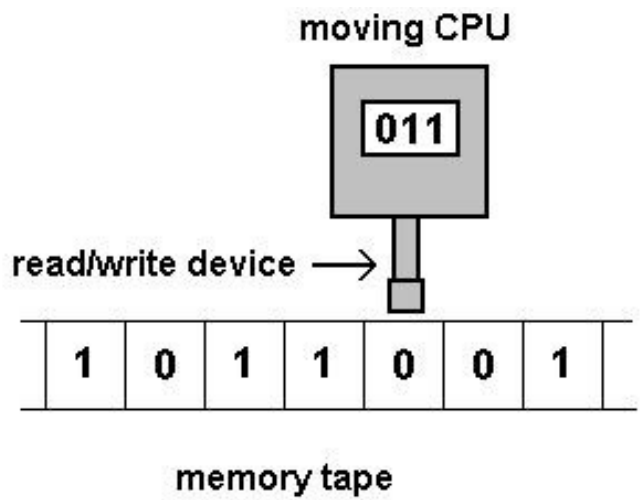


Figure: Turing Machine

All algorithms developed are based on Turing machine. It means algorithms are not written with particular CPU like Intel, AMD etc. in mind.

Turing Machine

It is mathematically proved that there is no mathematical/theoretical machine that is more computationally powerful (not considering efficiency of computation) than Turing machine.

It means it is possible to speed up real-life CPU but we can not create a CPU which can compute something that Turing machine cannot.

For example, Turing machine cannot solve “Halting problem”. It also means real-life CPU cannot solve “Halting problem” no matter how powerful CPU you use.

Turing Machine

It is not possible to create Turing machine in real-life as mathematical theory considers infinite memory which is not possible.

Turing machines are mathematically equivalent to each other. Modern CPUs like Intel, AMD, ARM are Turing complete which means they are mathematically equivalent to each other. It also means they are capable of running same algorithms.

Thank You ☺